

Bootstrapping

Fedora 12

If you've wanted to know what exactly goes on behind that cute boot splash when you boot Fedora, read on.

When you pop in the Fedora DVD bundled with this month's LFY, and install it on your computer, you may not feel anything has changed, except that the boot is kind of faster. Contrary to what you may think, it's not simply an optimisation of the *init* scripts that has made it faster — it's an entire change of architecture beneath that pretty boot splash screen. It's the first step to slash boot times to sub-15 seconds and yet make it a pretty, aesthetic and simple experience.

We try to peek into the intricate details of what goes on behind the scenes when we boot Fedora 12.

A system of Constantine proportions

A GNU/Linux system, in general, has three phases of booting: kernel and *initrd* load, system and then services initialisation. If that sounds gibberish, it basically means that in the first phase, the bootloader (GRUB

and its pals) loads the kernel and the initial RAM drive (*initrd*) into the memory. In the second phase, the essential initialisation tasks, like a filesystem check, UDEV device node generation, filesystem mounts, network initialisation, cryptographic services initialisation, setting the hostname, date and time and other basic stuff are done. And in the third phase, as the system enters a run-level, daemons and services like cups, httpd and the like are started.

The new architecture put in place in Fedora 12 brings three (relatively new and constantly improving) technologies under one umbrella, to tackle the second and third phases of the system boot. The system initialisation, which takes place mostly in the initial RAM drive, and partly just after the root filesystem has been mounted, is handled by a new tool called Dracut. A small part of the system initialisation and all the run-level and service management is done by another tool called Upstart, and to hide the intricacies and

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